



Ghazal Lahooti

Architectural & Urban Designer

Design Researcher · Climate & Human-Centred Designer

Evidence-Based Design · Computational Performance · Applied R&D

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🏠 Lund, Sweden

Profile

Architectural and urban designer/researcher working at the intersection of environmental performance, human-centred research, and computational design. Grounded in an MSc in Sustainable Urban Design and a background in Architectural Engineering, I approach design as an iterative, evidence-based process: *Research* → *Understand Context & People* → *Design* → *Evaluate* → *Iterate*. Rather than treating environmental analysis as an isolated post-design audit, I embed microclimate simulation, GIS spatial analytics, and qualitative user research directly into early-stage concept development to shape responsive spatial strategies. I am driven by R&D workflows that leverage scenario modelling, process automation, digital twins, and applied AI to connect rigorous inquiry with actionable, high-performance design.

Education

- **M.Sc. in Sustainable Urban Design**, Lund University, Sweden
Sep. 2024 – Sep. 2026 | GPA: Pass with Distinction (G/G)
Thesis: *GrönKöy: Strategic Destination Cooling and Age-Friendly Pedestrian Networks in Kadıköy, Istanbul*
Focus: Climate-adaptive urbanism, microclimate simulation, pedestrian thermal comfort, and demographic spatial analysis.
- **B.Sc. in Architectural Engineering**, Iran University of Science and Technology, Tehran, Iran
Sep. 2018 – Jul. 2023 | GPA: 16.95/20
Thesis: *Design of a Sustainable Mountain Refuge at the Third Campsite of Mount Damavand*
Focus: Passive climate strategies, extreme-environment envelope performance, and user-centric shelter design.

Design & Research Interests

- Climate-Adaptive Design & Outdoor Thermal Comfort
- Human-Centred Urbanism, Pedestrian Mobility & Demographic Analysis
- Performance-Driven Design & Multi-Objective Optimization
- GIS, Spatial Analytics & Urban Morphology
- Scenario Modelling, Alternative Testing & Decision-Support Systems
- Applied R&D: Computational Workflows, Digital Twins, Process Automation & Applied AI

Technical Skills & Methodologies

- **Design & 3D Modelling:** Rhino, Grasshopper, Revit, SketchUp, Blender, AutoCAD
- **Environmental Simulation & Performance:** Ladybug Tools (Ladybug, Honeybee), Eddy3D (Microclimate/CFD), Galapagos, Solar Radiation & Daylight Analysis
- **Spatial Analytics & Data Science:** QGIS, Google Earth Engine, Python (GeoPandas, Pandas, NumPy, Scikit-learn, Matplotlib)
- **Research & Analytical Methods:** PRISMA Systematic Reviews, Qualitative Stakeholder Interviews, Pedestrian Mobility/Destination Mapping, Spatial Demographic Analysis
- **Visualisation & Graphics:** Adobe Creative Suite (InDesign, Illustrator, Photoshop), D5 Render, Enscape

Research and Innovation Projects

- **GrönKöy (Master's Thesis): Strategic Destination Cooling and Age-Friendly Pedestrian Networks in Kadıköy, Istanbul**

Lund University | Supported by Crafoord Foundation Grant (SEK 30,000)

Investigated urban heat exposure and pedestrian mobility among older adults using GIS, remote sensing, and environmental simulation (Ladybug Tools). Formulated spatial cooling strategies centered on frequently visited public destinations and shared corridors rather than isolated residential streets, maximising the collective reach of heat mitigation interventions to ensure climate benefits are shared across diverse user groups.

- **UrbanSim (Personal Innovation Project / Concept Development) | SEK 40,000 Funding**

Sustainability Leapfrogs, LU Innovation | Summer 2026

Explores the development of a data-informed decision-support approach for context-specific urban heat adaptation through Nature-Based Solutions, investigating how spatial and environmental data can support the selection of appropriate cooling interventions across different urban conditions.

Research and Academic Experience

- **Project Assistant, Lund University and Malmö Municipality (3D Modelling and Solar Analysis)**
Oct. 2025 – Nov. 2025
 - Supported spatial modelling and solar radiation analysis for urban planning applications.
 - Contributed to data-driven evaluation of urban form and environmental performance.
- **Research Intern, Institute for Design Strategies, TH OWL University of Applied Sciences, Germany – AIM-NBS (Advancing Impact Measurement of Nature-Based Solutions)**
Jun. 2025 – Aug. 2025
 - Conducted PRISMA-based systematic literature review on impact assessment of Nature-Based Solutions.
 - Extracted and synthesised qualitative and quantitative indicators across 12 sustainability domains.

Professional Experience

- **Junior Architect, Cedrus Architecture Studio, Tehran**
Feb. 2023 – Aug. 2024
Contributed to site and environmental analysis, programme development, precedent research, and early-stage design for urban and housing projects.
- **Intern, Cedrus Architecture Studio, Tehran**
Jan. 2023 – Feb. 2023
Assisted senior architects in concept design, technical drawings, and client deliverables.

Awards and Honours

- **Leapfrogs Program Scholarship, LU Innovation:** SEK 40,000 awarded for the development of UrbanSim.
- **Crafoord Foundation Merit-Based Travel Grant:** SEK 30,000 awarded to support master's thesis research in Istanbul, Türkiye.
- **Lund University Merit-Based Global Scholarship (75% tuition):** Awarded for academic excellence.

Languages

- **Persian:** Native
- **English:** C1 (IELTS Academic 7.5)
- **German:** B1 (Goethe-Zertifikat), currently developing toward B2